

Raw-Wick heat-baseline orthogonality and rational current reduction

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claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1 Purpose and scope

R-100 wrote the regular coefficient martingale as $F_j = L_j + H_j$, paid the future-control residual H_j by one full-Wick R-094 secant allocation, and left the moving heat baseline L_j inside the complete covariance-normal row. This note identifies an exact centering which was missed in that ledger. It applies only to the isolated raw-Wick cross and therefore does not contradict R-100's warning about the complete row.

There are five advances.

1. For a pointwise coefficient, real-even Gaussian covariance, and a strict-past control, the moving heat coefficient L_j is conditionally orthogonal to the fresh full-Wick increment ΔQ_j .
2. Cross-Doob terminalization therefore reduces the complete terminal raw-Wick block to one fixed low endpoint plus exactly the R-100 future-control residual. A second R-094 charge would duplicate ownership.
3. The derivative-current increment has an exact raw-Wick plus current decomposition. Its control-only half telescopes to the terminal rational control square.
4. At the rational endpoint, the old R-085 shifted-Hessian-plus-square frontier is reduced to the strictly smaller current packet

$$\mathcal{K}_R = G^T Lc + \frac{1}{2} c^T B_1 c. \quad (1.1)$$

The $L : Q/2$ member is not estimated separately: it is already inside the complete raw-Wick block.

5. The production rational vector obeys an exact range identity and a pointwise norm contraction. This gives a useful Gram upper bound but no sign for a Wick-centred form.

The theorem is finite-cutoff and fixed-positive-floor, and its quantitative consequence uses the regular mutually orthogonal strict-past one-shot class. It does not prove a lower bound for $\mathcal{K}_R$, arbitrary progressive or revisit controls, complete rational H_N , REG, H_A , OVERLAP_{src}, the $q = 10/9$ Nelson estimate, cutoff or floor removal, an interacting measure, or Sector A. Tier stays T4.

2 Finite-cutoff setting and hypotheses

Let g_j be independent centred Gaussian value roots and let $d_{j,i} = D_i g_j$ be the derivative of the j th root in direction i . Write

$$U_j = \sum_{s \leq j} g_s, \quad G_{j,i} = D_i U_j = \sum_{s \leq j} d_{s,i}. \quad (2.1)$$

With

$$\Delta \Gamma_{j,i} = \text{Cov}(d_{j,i}), \quad \Gamma_{j,i} = \sum_{s \leq j} \Delta \Gamma_{s,i}, \quad (2.2)$$

define the quadratic Wick martingale

$$Q_{j,i} = G_{j,i}^{\otimes 2} - \Gamma_{j,i}. \quad (2.3)$$

Its increment is

$$\Delta Q_{j,i} = 2G_{j-1,i} \odot d_{j,i} + (d_{j,i}^{\otimes 2} - \Delta \Gamma_{j,i}). \quad (2.4)$$

Thus $\mathbb{E}_{j-1} \Delta Q_{j,i} = 0$.

The regulator is real and even. At every point x this gives

$$\text{Cov}(g_j(x), d_{j,i}(x)) = 0. \quad (2.5)$$

Joint Gaussianity turns (2.5) into pointwise independence. This is a same-point statement; $g_j(y)$ and $d_{j,i}(x)$ need not be independent when $x \neq y$.

Let

$$a_j \in \mathcal{F}_{j-1}, \quad A^{(j)} = A^0 + \sum_{j_0 \leq s \leq j} a_s, \quad A^* = A^{(J)}, \quad (2.6)$$

and put

$$Z_* = U_J + A^*, \quad Z_j = U_J + A^{(j)}. \quad (2.7)$$

For a pointwise matrix coefficient $B : \mathbb{R}^N \rightarrow \text{Sym}_N$ of sufficient polynomial growth, define

$$F_j = \mathbb{E}_j B(Z_*), \quad (2.8)$$

$$L_j = \mathbb{E}_j B(Z_j) = P_{\Sigma[j+1:J]} B(U_J + A^{(j)}), \quad (2.9)$$

and

$$H_j = \mathbb{E}_j [B(Z_*) - B(Z_j)]. \quad (2.10)$$

Then $F_j = L_j + H_j$ exactly. For matrix fields, set

$$\langle C, Q \rangle = \sum_i \int C(x) : Q_i(x) \, dx. \quad (2.11)$$

The exact theorem uses finite cutoff, conditional Fubini, independent Gaussian roots, (2.5), strict-past measurability in (2.6), and spatial locality of $B(Z)(x)$ in the value $Z(x)$. The R-100 quantitative residual bound additionally uses regular mutually orthogonal one-shot controls.

3 Moving raw-Wick heat-baseline orthogonality

Theorem 3.1 (fresh value–gradient orthogonality)

For every j and derivative direction i ,

$$\boxed{\mathbb{E}_{j-1} \langle L_j, \Delta Q_{j,i} \rangle = 0.} \quad (3.1)$$

Consequently,

$$\boxed{\mathbb{E} \langle L_j - L_{j-1}, \Delta Q_{j,i} \rangle = 0.} \quad (3.2)$$

Proof. Fix x and condition on $\mathcal{F}_{j-1}$. Then $U_{j-1}(x)$, $A^{(j)}(x)$, and $G_{j-1,i}(x)$ are fixed. There is a deterministic matrix-valued function

$$\Phi_{j,x}(v) = P_{\Sigma[j+1:J]}B(U_{j-1}(x) + A^{(j)}(x) + v) \quad (3.3)$$

such that $L_j(x) = \Phi_{j,x}(g_j(x))$. By (2.5), $g_j(x)$ is independent of $d_{j,i}(x)$. Hence

$$\begin{aligned} \mathbb{E}_{j-1}[L_j(x) : 2G_{j-1,i}(x) \odot d_{j,i}(x)] &= 0, \\ \mathbb{E}_{j-1}[L_j(x) : (d_{j,i}(x)^{\otimes 2} - \Delta\Gamma_{j,i})] &= 0. \end{aligned} \quad (3.4)$$

Equation (2.4) and conditional spatial Fubini prove (3.1). Since L_{j-1} is $\mathcal{F}_{j-1}$ -measurable,

$$\mathbb{E}\langle L_{j-1}, \Delta Q_{j,i} \rangle = \mathbb{E}\langle L_{j-1}, \mathbb{E}_{j-1} \Delta Q_{j,i} \rangle = 0, \quad (3.5)$$

which proves (3.2). $\square$

The scope is exact and narrow. The theorem centres the isolated cross $\langle d_j L, d_j Q \rangle$. It does not centre the complete moving covariance-normal row, which also contains control derivatives, the terminal square, heat and trace corrections, the fixed-low subtraction, payment, and the R-063 reconstruction forest. R-100's broader warning remains correct for that complete row; its phrase “no standalone centering” is superseded only for the raw-Wick cross (3.2).

The hypotheses are necessary in the following sense. If value and gradient coordinates are correlated, a coefficient quadratic in the value has a nonzero cross with the centred derivative square. If the coefficient is spatially nonlocal or depends directly on the fresh derivative, the factorisations in (3.4) are unavailable. Same-root and revisit controls are also outside (2.6).

4 Cross-Doob terminalization without duplicate payment

Theorem 4.1 (terminal raw-Wick identity)

For every starting root $\ell < J$,

$$\boxed{\mathbb{E}\langle B(Z_*), Q_J \rangle = \mathbb{E}\langle F_\ell, Q_\ell \rangle + \sum_{j>\ell} \mathbb{E}\langle B(Z_*) - B(Z_j), \Delta Q_j \rangle.} \quad (4.1)$$

Proof. Both $F_j = \mathbb{E}_j B(Z_*)$ and Q_j are martingales. Their cross-Doob identity is

$$\mathbb{E}\langle F_j, Q_j \rangle - \mathbb{E}\langle F_{j-1}, Q_{j-1} \rangle = \mathbb{E}\langle F_j - F_{j-1}, \Delta Q_j \rangle. \quad (4.2)$$

Sum (4.2), write $F = L + H$, and use Theorem 3.1 on the L increment. The H_{j-1} cross vanishes by prediction, while ΔQ_j is $\mathcal{F}_j$ -measurable, so

$$\mathbb{E}\langle H_j, \Delta Q_j \rangle = \mathbb{E}\langle B(Z_*) - B(Z_j), \Delta Q_j \rangle. \quad (4.3)$$

Finally $F_J = B(Z_*)$. $\square$

Take $\ell = j_0 - 1$. R-100 Theorem 5.1 proves

$$\sum_{j=j_0}^J \left| \mathbb{E} \int [B(Z_*) - B(Z_j)] : \Delta Q_j \, dx \right| \leq C_{\text{full}} 2^{-j_0} X^{1/2} (1 + Y)^{1/6}, \quad (4.4)$$

and hence, for every $\eta, \zeta \in (0, 1]$,

$$|\mathbb{E}\langle B(Z_*), Q_J \rangle - \mathbb{E}\langle F_{j_0-1}, Q_{j_0-1} \rangle| \leq \eta \mathbb{E}X + \zeta \mathbb{E}Y + C_{\eta, \zeta}. \quad (4.5)$$

Thus the entire terminal raw-Wick coefficient is assigned once: its fixed low endpoint is retained and its future-control reveal is paid by (4.4). No Taylor component of $B(Z_*) : Q_J$ is paid separately after (4.5). In particular, applying another R-094 secant to a shifted $L : Q$ component would double count the same raw-Wick owner.

5 Exact derivative-current remainder

Let B_- and B_+ be symmetric matrices at two consecutive coefficient endpoints, let G be the Gaussian derivative prefix, let C be the old control derivative, let b be its fresh strict-past increment, and let Γ be the Gaussian derivative covariance. Put

$$Q = G^{\otimes 2} - \Gamma, \quad \Delta B = B_+ - B_-. \quad (5.1)$$

Lemma 5.1 (raw-Wick plus current partition)

The full quadratic increment obeys

$$\begin{aligned} & \frac{1}{2} \left\{ B_+ : ((G + C + b)^{\otimes 2} - \Gamma) - B_- : ((G + C)^{\otimes 2} - \Gamma) \right\} \\ &= \frac{1}{2} \Delta B : Q + K, \end{aligned} \quad (5.2)$$

where

$$K = G^T \Delta B C + \frac{1}{2} C^T \Delta B C + (G + C)^T B_+ b + \frac{1}{2} b^T B_+ b. \quad (5.3)$$

Moreover,

$$\boxed{K = G^T \{B_+(C + b) - B_- C\} + \frac{1}{2} \{(C + b)^T B_+(C + b) - C^T B_- C\}.} \quad (5.4)$$

Proof. Expand both symmetric quadratic forms. The $G^{\otimes 2} - \Gamma$ terms give $\Delta B : Q/2$; collecting the remaining mixed and control-only monomials gives (5.3). Expanding (5.4) reproduces (5.3) term by term. $\square$

The square telescope is an expectation identity across different heat levels, not a pathwise identification. Fix the mandatory target heat Σ_{tar} and put

$$\overline{B}_k^{(J)} = P_{\Sigma_{\text{tar}} + \Sigma_{k+1:J}} B, \quad C^{(k)} = D A^{(k)}. \quad (5.5)$$

For a predictable insertion, $A^{(k)}, C^{(k)} \in \mathcal{F}_{k-1}$, and heat composition gives the exact conditional matching

$$\begin{aligned} \mathbb{E}_{k-1} [\overline{B}_k^{(J)}(U_k + A^{(k)})] &= P_{\Delta \Sigma_k} P_{\Sigma_{\text{tar}} + \Sigma_{k+1:J}} B(U_{k-1} + A^{(k)}) \\ &= \overline{B}_{k-1}^{(J)}(U_{k-1} + A^{(k)}). \end{aligned} \quad (5.6)$$

Since $C^{(k)}$ is predictable, multiplication by its quadratic form and conditional Fubini show that the positive endpoint at level k equals the matching negative endpoint in expectation. Consequently the control-square parts of (5.4) telescope to

$$\begin{aligned} & \frac{1}{2} \mathbb{E} \sum_i \int (C_i^{(J)})^T (P_{\Sigma_{\text{tar}}} B) (U_J + A^*) C_i^{(J)} dx \\ & - \frac{1}{2} \mathbb{E} \sum_i \int (C_i^{(j_0-1)})^T \overline{B}_{j_0-1}^{(J)} (U_{j_0-1} + A^{(j_0-1)}) C_i^{(j_0-1)} dx. \end{aligned} \quad (5.7)$$

The fixed-low term is retained unless its control derivative is zero; the terminal member is nonnegative. The only signed new part of (5.4) is therefore the current cross

$$G^T \{B_+(C + b) - B_- C\}. \quad (5.8)$$

This algebra does not assert that (5.8) has a sign.

6 Rational endpoint and the smaller K -frontier

At the reassembled rational endpoint put

$$\bar{B} := \bar{B}_{\Sigma_{\text{tar}}} = P_{\Sigma_{\text{tar}}} B_R, \quad B_0 = \bar{B}(U), \quad B_1 = \bar{B}(U + a), \quad G = DU, \quad c = Da, \quad (6.1)$$

and $Q = G^{\otimes 2} - \Gamma$. Define

$$K_t = D^2 \bar{B}(U + ta) - D^2 \bar{B}(U), \quad (6.2)$$

$$L = \int_0^1 (1 - t) K_t[a, a] dt. \quad (6.3)$$

Taylor's theorem gives both exact forms

$$L = B_1 - B_0 - D\bar{B}(U)[a] - \frac{1}{2} D^2 \bar{B}(U)[a, a], \quad (6.4)$$

$$L = \frac{1}{2} \int_0^1 (1 - t)^2 D^3 \bar{B}(U + ta)[a, a, a] dt. \quad (6.5)$$

At fixed positive floor, differentiation commutes with the deterministic target heat:

$$D^m \bar{B} = P_{\Sigma_{\text{tar}}} D^m B_R. \quad (6.5a)$$

Set

$$B_T = B_0 + D\bar{B}(U)[a] + \frac{1}{2} D^2 \bar{B}(U)[a, a], \quad (6.6)$$

so $B_1 = B_T + L$.

Theorem 6.1 (exact nonduplicating endpoint ledger)

The rational endpoint increment

$$\Delta \mathcal{W}_R = \frac{1}{2} (B_1 - B_0) : Q + G^T B_1 c + \frac{1}{2} c^T B_1 c \quad (6.7)$$

has the exact decomposition

$$\boxed{\Delta \mathcal{W}_R = \mathcal{R}_Q + \mathcal{M}_U + \mathcal{K}_R,} \quad (6.8)$$

where

$$\mathcal{R}_Q = \frac{1}{2} (B_1 - B_0) : Q, \quad (6.9)$$

$$\begin{aligned} \mathcal{M}_U = G^T B_T c &= G^T B_0 c + G^T D\bar{B}(U)[a]c \\ &\quad + \frac{1}{2} G^T D^2 \bar{B}(U)[a, a]c, \end{aligned} \quad (6.10)$$

and

$$\boxed{\mathcal{K}_R = G^T L c + \frac{1}{2} c^T B_1 c = \int_0^1 (1 - t) G^T K_t[a, a] c dt + \frac{1}{2} c^T B_1 c.} \quad (6.11)$$

Proof. Insert $B_1 = B_T + L$ into the mixed term of (6.7), while keeping the raw-Wick difference intact. $\square$
R-085 used

$$\mathcal{H}_R = \frac{1}{2}L : Q + G^T Lc \quad (6.12)$$

and left the frontier

$$F_{6.5} = \mathcal{H}_R + \frac{1}{2}c^T B_1 c = \frac{1}{2}L : Q + \mathcal{K}_R. \quad (6.13)$$

The new ledger does not estimate $L : Q/2$ alone. Equation (6.9) retains

$$\frac{1}{2} \left(D\overline{B}(U)[a] + \frac{1}{2}D^2\overline{B}(U)[a, a] + L \right) : Q \quad (6.14)$$

as one raw-Wick block. Apply Theorem 4.1 to the local coefficient $P_{\Sigma_{\text{tar}}} B_R$: it assigns the complete terminal $B_1 : Q/2$ member to the fixed low endpoint and R-100 residual, while the unshifted $B_0 : Q/2$ member remains in the accepted base Wick/forest owner. The first two old covariance-normal Taylor families must therefore not be charged again. R-071/R-085 pays the three unshifted current families in (6.10). The honest remaining rational packet is $\mathcal{K}_R$, not $F_{6.5}$.

Complete-owner row additivity is essential. The three linear Pauli–Fierz rows have quadratic Gram matrices, hence $K_t = 0$, and their complete endpoint is already paid by R-084. Only the rational row contributes (6.11).

7 Production rational range identity

Let P_{dbl} be the real doublet projector and write

$$r = |P_{\text{dbl}}z|^2, \quad d = |z|^2 + \epsilon_\rho, \quad \alpha = \frac{5}{9}. \quad (7.1)$$

The rational row vector and Gram matrix are

$$g(z) = P_{\text{dbl}}z - \alpha \frac{r}{d}z, \quad B_R(z) = 4c_1 g(z) \otimes g(z), \quad (7.2)$$

with

$$c_0 = \frac{3}{250P}, \quad c_1 = \frac{243}{8000P}, \quad P = 4 + 10^{-12}. \quad (7.3)$$

These constants follow exactly from

$$Q_{\text{II}} = \begin{pmatrix} 9/(500P) & 3/(400P) \\ 3/(400P) & 3/(320P) \end{pmatrix}. \quad (7.4)$$

Lemma 7.1 (range contraction)

For every z and positive floor,

$$\boxed{r - |g(z)|^2 = \alpha \frac{r^2}{d^2} \{2d - \alpha|z|^2\} \geq 0.} \quad (7.5)$$

Consequently,

$$|g(z)| \leq |P_{\text{dbl}}z|, \quad 0 \leq B_R(z) \leq 4c_1 r I. \quad (7.6)$$

After target heat this gives only the upper estimate

$$0 \leq \overline{B}(z) \leq 4c_1 \left(|P_{\text{dbl}}z|^2 + \text{tr}(P_{\text{dbl}}\Sigma_{\text{tar}}) \right) I. \quad (7.7)$$

Proof. Use $(P_{\text{dbl}}z) \cdot z = r$ to expand $|g|^2$. The bracket in (7.5) is nonnegative because $0 < \alpha < 2$ and $d > |z|^2$. The rank-one inequality $g \otimes g \preceq |g|^2 I$ proves (7.6). $\square$

These are upper-range statements. Wick centering subtracts covariance and does not preserve point-wise matrix order, while an upper coefficient bound does not supply lower coercivity, so (7.6)–(7.7) do not prove the signed $\mathcal{K}_R$ lower form.

8 The exact remaining obstruction

The three unshifted families and the raw-Wick block are already assigned. Taylor’s theorem and (6.5a) split

$$L = \frac{1}{6} D^3 \bar{B}(U)[a, a, a] + L_{\text{bal}},$$

where

$$L_{\text{bal}} = \int_0^1 (1-t) \int_0^t \{D^3 \bar{B}(U + sa) - D^3 \bar{B}(U)\}[a, a, a] ds dt.$$

The base-frozen current $G^T D^3 \bar{B}(U)[a, a, a]c/6$ is assigned to the R-076 smooth geometric one-form model. The unresolved heat-lifted balanced current is

$$\mathcal{H}_{\text{bal}} = \sum_i \int \int_0^1 (1-t) \int_0^t G_i^T \{D^3 \bar{B}(U + sa) - D^3 \bar{B}(U)\}[a, a, a] c_i ds dt dx. \quad (8.1)$$

Thus a sharp sufficient successor is the following, which is a target rather than a theorem of R-101: for fixed $0 < \theta < 1$, uniformly over the admissible target-heat family and terminal cutoff,

$$\boxed{\mathbb{E} \mathcal{H}_{\text{bal}} + \frac{\theta}{2} \mathbb{E} \sum_i \int (D_i a)^T \bar{B}(U + a) D_i a dx \geq -\eta \mathbb{E} X - \zeta \mathbb{E} Y - C_{\eta, \zeta, \theta, e, j0}.} \quad (8.2)$$

The constant in (8.2) is independent of the admissible Σ_{tar} and of the terminal cutoff. Equivalently, a formulation with fixed-heat constants must prove their uniform supremum finite over the production-admissible heat family. Equivalently, for an independent centred target-heat variable $h_{\Sigma_{\text{tar}}}$,

$$(D_i a)^T \bar{B}(U + a) D_i a = \mathbb{E}_{h_{\Sigma_{\text{tar}}}} |C_R(U + a + h_{\Sigma_{\text{tar}}}) D_i a|^2. \quad (8.2a)$$

It is not generally the unheated square $|C_R(U + a) D_i a|^2$. A positive $(1 - \theta)$ fraction of the same heat-lifted terminal square is retained for later assembly. Equation (8.2) is the heat-lifted coefficient-balanced rational-current Carleson gate.

The present authorities do not prove (8.2).

1. R-096 removes coefficient outputs strictly above the control shell by a fixed gap. The balanced $m = k + O(1)$ high-high-to-low branch survives.
2. Absolute input-maximum control produces $X^{2/5} Y^{7/10}$, whose total power is $11/10 > 1$.
3. Separating the shifted multiplier at the accepted $s = 3/5$ produces the supercritical total

$$\frac{5+s}{6} + \frac{s}{4} = \frac{13}{12}. \quad (8.3)$$

4. A shell-dependent square fraction cannot repair both the inverse Schur cost and the forward square telescope simultaneously.

A square-only pointwise Schur proof cannot be uniform down to zero target heat. At $\Sigma_{\text{tar}} = 0$, on the scalar rational ray with initial value $\sqrt{e}$ and endpoint approaching zero, the cubic Taylor remainder remains nonzero while the endpoint Gram vanishes quadratically. For scalar target variance $\sigma \downarrow 0$ at the endpoint zero,

$$(P_\sigma B_R)(0) = 4c_1\sigma + O(\sigma^2/e), \quad L_\sigma = \frac{40}{81}c_1e + O(\sigma),$$

but this is the full second-order remainder. The already-assigned base cubic and the actual unresolved balanced remainder are

$$\frac{1}{6}(P_\sigma B_R)'''(\sqrt{e})(-\sqrt{e})^3 = -\frac{25}{81}c_1e + O(\sigma), \quad L_{\text{bal},\sigma} = \frac{65}{81}c_1e + O(\sigma).$$

Consequently the gate-specific optimised scalar square completion is

$$-\frac{4225}{52488\theta} \frac{c_1e^2}{\sigma} G^2 + O(1),$$

and diverges to negative infinity. At a fixed strictly positive full-rank heat the convolution can fill the kernel, and no fixed-heat no-go is asserted. The zero-heat-uniform obstruction is still not a counterexample to (8.2), because the spatial Cameron–Martin term may pay the derivative of the control.

There is a second, independent diagnostic route. For one paired real mode and the full mandatory future heat, the quadratic control coefficient can be written exactly as a Gaussian expectation of rational derivatives of g . Laplace representation of the denominator reduces it to a deterministic one-dimensional integral of a finite Wick polynomial against a tilted Gaussian covariance. This supplies a viable interval-plus-asymptotic route, but no all-mode sign theorem has yet been proved. Numerical sampling or a finite list of modes is not used as evidence for R-101.

9 Failed routes, corrections, and proof map

Route	Verdict	Reusable conclusion
Moving L_j has no raw centering	corrected	The complete row has no standalone sign, but Theorem 3.1 centres its isolated raw-Wick cross exactly.
Second secant on shifted $L : Q$	failed	It duplicates the complete raw-Wick owner already terminalized by Theorem 4.1.
Clark–Ocone alone	failed	It rewrites the cross as an innovation bracket without supplying a spatial or root weight.
Gaussian integration by parts	failed	Differentiating adapted future controls introduces uncontrolled first and second Malliavin derivatives.
Hermite ranks zero boundary through two	boundary	They identify raw-Wick ownership but supply no spatial gain for the current cross.
Zero-heat-uniform end-point Schur	boundary	The unheated rational Gram has kernel leakage, but a fixed positive full-rank heat can fill that kernel; no uniform constant as target heat tends to zero follows.
Absolute balanced summation	failed	The powers in (8.3) are supercritical.
Full future-heat mode reduction	inconclusive	Exact one-dimensional Laplace–Wick certification is available; the uniform analytic tail is open.

The nonduplicating evidence path is

$$\begin{aligned}
& \text{R-063 forest and fixed low block} \\
& \longrightarrow \text{R-097 heat matching} \\
& \longrightarrow \text{R-100 full-Wick future residual} \\
& \longrightarrow \text{R-101 heat-baseline orthogonality} \\
& \longrightarrow \text{R-084 linear rows plus R-071/R-076 currents} \\
& \longrightarrow \boxed{\text{heat-lifted } \mathcal{K}_R \text{ / gate (8.2)}} \\
& \longrightarrow H_N \longrightarrow \text{REG} \longrightarrow H_A \longrightarrow \text{OVERLAP}_{\text{src}} \longrightarrow \text{Nelson and measure construction.}
\end{aligned} \tag{9.1}$$

The exploration ledger records the decisions as EXP-000243–EXP-000249. EXP-000248 preserves the target-heat placement correction, while EXP-000249 separates the total 40/81 remainder from the actual balanced 65/81 successor and enforces target-heat uniformity. These records continue and correct EXP-000238 and EXP-000242 without deleting historical decisions.

Proof-status map

Object	Status after R-101	Next condition
Moving raw-Wick heat base-line	closed in regular strict-past class	Keep locality, value–gradient independence, low block, and R-100 residual once.
Three linear production rows	accepted from R-084	Do not reuse their complete endpoint companions.
Rational raw-Wick block	reduced	Book only through Theorem 4.1 and the accepted base/low authority.
Rational unshifted current	accepted from R-071/R-085	Three families in (6.10), not the two raw Taylor families.
Rational shifted current	open	Prove heat-lifted coefficient-balanced gate (8.2).
Complete regular H_N and REG	open	Reassemble only after (8.2).
Progressive/revisit H_A	open	Lossless extension beyond strict-past one-shot remains separate.
OVERLAP _{src} , Nelson, measure	open	Use R-087 CORE only after packet and one-use assembly.
Sector A	open	No closure or tier promotion.

10 Devil’s-advocate review

Objection 1: L_j and ΔQ_j use the same Gaussian root. DISMISSED under the hypotheses. At a fixed spatial point, L_j uses the fresh value coordinate and ΔQ_j uses the fresh derivative coordinate. Real-even covariance and Gaussianity give independence. **VALID outside the stated scope.** A nonlocal coefficient can see $g_j(y)$ correlated with $d_j(x)$.

Objection 2: a same-root control can be admitted by approximation. UPHELD as false. If a_j sees the fresh root, (3.3) need not be a function of $g_j(x)$ alone after conditioning. The theorem is strict-past and does not claim progressive density.

Objection 3: Theorem 4.1 proves the complete owner positive. UPHELD as false. It controls only the terminal raw-Wick block. The mixed derivative current, terminal control square, base, payment, heat, trace, and forest owners remain in the ledger.

Objection 4: the first two R-085 Taylor families may still be used as an extra reserve. UPHELD as double counting. They are constituents of the complete raw block (6.14), which has already been terminalized and paid once.

Objection 5: positivity of B_1 controls $G^T Lc$ uniformly. UPHELD as unsupported. At fixed strictly positive target heat, $B_1 = P_{\Sigma_{\text{tar}}} B_R(U + a)$ can have better rank than the unheated Gram. Square completion would nevertheless require a cutoff- and root- uniform estimate of $G^T L B_1^\dagger L^T G$ together with range control. No current authority supplies it. The exact unheated fixture proves only that such an estimate cannot be uniform as $\Sigma_{\text{tar}} \downarrow 0$.

Objection 6: the unheated range contraction orders the heat-lifted Wick form. UPHELD as false. Heat averaging gives only the upper bound (7.7). Subtracting covariance is not monotone under coefficient PSD order, and an upper coefficient bound does not prove the missing signed inequality.

Objection 7: the target square may be written without $P_{\Sigma_{\text{tar}}}$. UPHELD as false. The endpoint coefficient in Section 6 is $B_1 = P_{\Sigma_{\text{tar}}} B_R(U + a)$. Its square is the heat expectation (8.2a). Dropping this heat changes the theorem.

Objection 8: the finite-filtration executables prove the analytic theorem by enumeration. DISMISSED. The proof is (3.3)–(3.5) and (4.2)–(4.3). The executables are adversarial certificates for the algebra, scope failures, Taylor ownership, and production constants; they are not a substitute for the conditional-Fubini proof.

Objection 9: this finishes Sector A. UPHELD as false. Gate (8.2), complete H_N , REG, progressive assembly, OVERLAP_{src}, Nelson, cutoff/floor removal, and the measure theorem remain open. Tier is T4 before and after.

Executable adversarial audit. Sign and factor checks retain the physical outer $1/2$, symmetric mixed contraction, and six-real $B_R = 4c_1 g \otimes g$ convention. The target square uses $P_{\Sigma_{\text{tar}}}$, not an unheated substitute. Production constants and floor are derived from the hash-pinned A1 manifest; the independent route parses the same declared decimals into exact fractions without importing the primary code. Limit fixtures include a nonzero fixed-low endpoint, correlated value-gradient roots, a derivative-dependent coefficient, a fresh-root control, a two-site nonlocal coefficient, zero and near-floor rational states, pure doublet and kernel states, and generic states. No convergence as $J \rightarrow \infty$, $\epsilon_\rho \downarrow 0$, or $\Sigma_{\text{tar}} \downarrow 0$ is asserted. External review is invited on heat placement, the real/complex covariance factor, the endpoint-square telescope, the balanced-versus-total Taylor remainder, target-heat uniformity, and the no-double-counting ledger.

11 Executable verification

The primary verifier checks a nonlinear three-root filtration, a genuinely nonzero fixed-low endpoint, scope-breaking fixtures, random noncommuting derivative-current identities, the matched control-square telescope, exact Taylor ownership, constants and floor derived directly from the A1 manifest, and zero, near-floor, doublet, kernel, and generic rational range cases. The independent verifier uses only the Python standard library, parses A1 decimals as exact fractions, and imports neither the primary verifier nor NumPy. The two routes independently derive and assert the scalar coefficients $40/81$, $-25/81$, $65/81$, 4 , and $4225/52488$. The integrated verifier reruns both, cross-checks their five production constants, pins upstream and package hashes, audits the proof note and PDF, and checks all public surfaces, count

contracts, and no-overclaim flags. The final contracts are primary 187/187, independent 38/38, integrated 276/276, and aggregate 501/501.

```
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/  
a13_classii_raw_wick_heat_baseline_orthogonality_  
rational_current_reduction_verify.py
```

Result footer

Result ID: A13-CLASSII-RAW-WICK-HEAT-BASELINE-ORTHOGONALITY-
RATIONAL-CURRENT-REDUCTION

Ledger: R-101.

Title: Raw-Wick heat-baseline orthogonality and rational
current reduction.

Precise statement:

Under locality, real-even same-point value-gradient covariance,
and strict-past control, the moving heat coefficient is
conditionally orthogonal to the fresh full-Wick increment.
Cross-Doob terminalization assigns the complete heat-lifted raw
block once. The remaining regular rational packet is
 $K_R = G^T L c + c^T (P_{\text{Sigma}} B_R) (U + a) c / 2$.

Scope: Finite cutoff, fixed positive floor, local coefficient; regular
mutually orthogonal strict-past one-shot controls for the
quantitative residual. No same-root or revisit extension.

Dependencies:

A1 production-functional manifest; R-063; R-071; R-076; R-083;
R-084; R-085; R-094; R-096; R-097; R-100.

Evidence grade:

ANALYTIC, EXACT, EXECUTED, BOUNDARY.

Reproduction command:

```
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/  
a13_classii_raw_wick_heat_baseline_orthogonality_  
rational_current_reduction_verify.py
```

Expected output:

Manifest-pinned primary, standard-library independent,
integrated, and aggregate PASS; exit zero.

Explorations:

EXP-000243--EXP-000249.

Falsification gate:

Failure of locality/value-gradient independence/strict-past
scope, P_{Sigma} heat composition, the indexed square telescope,
cross-Doob identity, heat-lifted Taylor ownership, range identity,
upstream hashes, PDF QA, ledger/count, or T4 boundary.

Tier before / after:

T4 / T4; no promotion.

No-overclaim statement:

No lower bound for the heat-lifted balanced K_R packet, no
fixed-heat coercivity theorem, no uniform $\text{Sigma} > 0$ statement,
complete H_N , REG, progressive H_A , OVERLAP_src, Nelson,
cutoff/floor removal, measure construction, T5--T7, or Sector-A
closure is proved.

Next required action:

Prove (8.2) uniformly over the admissible target heats with the
actual P_{Sigma} endpoint square, retaining a positive terminal-
square fraction, before regular H_N/REG reassembly and the
progressive R-087 CORE/Nelson bridge.